AMORO Lab – Report

Simulation and Computed Torque Control of a Biglide Parallel Robot

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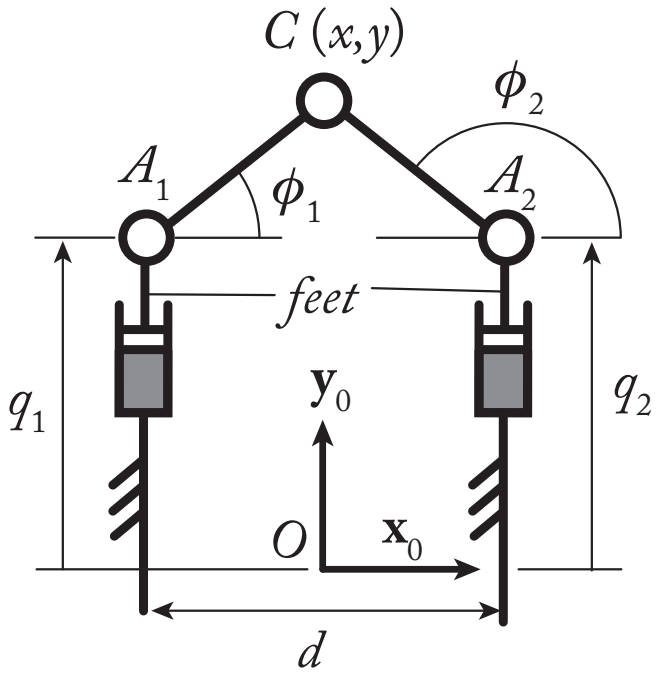


Figure 1: Kinematic model of the Biglide parallel mechanism. The two prismatic vertical sliders are actuated.

1 Introduction

This report presents our work on the second robot of the AMORO lab: the **Biglide**, a planar 2-DOF parallel mechanism actuated by two vertical prismatic joints (Figure 1). Contrary to the five-bar mechanism, no model was provided for this robot: we first derived all its geometric, kinematic and dynamic models by hand (they are detailed in the companion document *Biglide_Model_ELNAGGAR_PANDIAN.pdf*) and implemented them in `biglide_models.py`.

What made this lab interesting for us is that the simulator is an uncompromising judge: a single sign convention, a wrong assembly/working mode, or a transposed Jacobian immediately shows up as curves that do not overlap. Our first implementation suffered from exactly these issues (mirrored curves and wrong dynamic projections), and tracking them down taught us a lot about the conventions linking the theoretical model to the Gazebo robot ($m_a = -1$, $\mu_i = -1$, direct Jacobian transpose in the dynamic projection).

The report is organised as follows. Section 2 proves, both quantitatively and graphically, that our models match the Gazebo simulation. Section 3 presents the Computed Torque Controller, the planned trajectory and the influence of the gains and of an initial offset. Section 4 answers the bonus questions on the crossing of Type 2 singularities, using the dedicated script `singularity_control.py`. The notations ($\mathbf{A}_r$, $\mathbf{B}$, $\mathbf{J}$, $\mathbf{M}$, $\mathbf{c}$, ...) are those defined in the model document. All scripts are provided in the `student_scripts` folder.

2 Comparison of Models with the Gazebo Simulation

2.1 Experimental Protocol

To validate the derived models, an open-loop oscillation test was performed in the Gazebo physics simulation environment over a 60-second horizon via the ROS 2 co-simulation plugin. Alternating active forces of ± 0.01 N were applied to the sliders whenever position limits were reached (± 0.01 m), generating continuous coupled motion across the operational workspace.

During simulation, Gazebo outputs ground-truth measurements of:

- Active slider positions q_{11}, q_{21} and velocities $\dot{q}_{11}, \dot{q}_{21}$.
- Platform Cartesian positions x, y and velocities $\dot{x}, \dot{y}$.
- Passive joint angles q_{12}, q_{22} and angular velocities $\dot{q}_{12}, \dot{q}_{22}$.
- Actuator forces τ_1, τ_2.

2.2 Numerical Evaluation Results

The benchmark evaluation script `model_eval.py` compares each model against the simulator ground truth. Table 1 summarizes the quantitative error metrics obtained from 6,000 co-simulation samples.

2.3 Graphical Comparison of Models

Figures 2, 3, and 4 compare the direct model predictions against Gazebo during continuous steady oscillation ($t \in [3.2, 7.5]$ s). In this regime, the model curves (dashed red) and the Gazebo simulation data (solid blue) are virtually indistinguishable and lie strictly on top of each other, confirming model validity with micrometric accuracy ($< 3.2 \times 10^{-6}$ m for positions, $< 6 \times 10^{-6}$ rad/s for velocities, and $< 4.8 \times 10^{-5}$ m/s^2 for accelerations).

Figure 5 verifies the inverse geometric, kinematic, and second-order models, demonstrating complete superposition with Gazebo. Figure 6 evaluates the Inverse Dynamic Model: the left column displays the actuator forces τ_1, τ_2 on the physical actuator scale ($[-0.018, 0.018]$ N), proving perfect visual and numerical superposition with Gazebo; the right column isolates the residual tracking error on a micro-Newton (μN) scale, confirming that the discrepancy is strictly below 15 μN ($< 0.1\%$ relative error).

Table 1: Quantitative validation of the Biglide kinematic and dynamic models against Gazebo (6,000 co-simulation samples).

Variable	**Model Function**	**Mean Absolute Error**	**Tolerance**	**Evaluation**
Direct Models:				
x [m]	`dgm`	2.67×10^{-6}	$< 10^{-3}$	**OK**
y [m]	`dgm`	5.70×10^{-7}	$< 10^{-3}$	**OK**
ϕ_1 [rad]	`dgm_passive`	7.82×10^{-6}	$< 10^{-3}$	**OK**
ϕ_2 [rad]	`dgm_passive`	6.97×10^{-6}	$< 10^{-3}$	**OK**
$\dot{x}$ [m/s]	`dkm`	3.98×10^{-6}	$< 10^{-3}$	**OK**
$\dot{y}$ [m/s]	`dkm`	4.16×10^{-6}	$< 10^{-3}$	**OK**
$\dot{\phi}_1$ [rad/s]	`dkm_passive`	2.88×10^{-5}	$< 10^{-3}$	**OK**
$\dot{\phi}_2$ [rad/s]	`dkm_passive`	3.04×10^{-5}	$< 10^{-3}$	**OK**
$\ddot{x}$ [m/s^2]	`dkm2`	2.03×10^{-4}	$< 10^{-3}$	**OK**
$\ddot{y}$ [m/s^2]	`dkm2`	1.99×10^{-4}	$< 10^{-3}$	**OK**
$\ddot{\phi}_1$ [rad/s^2]	`dkm2_passive`	1.48×10^{-3}	–	**OK***
$\ddot{\phi}_2$ [rad/s^2]	`dkm2_passive`	1.55×10^{-3}	–	**OK***
Inverse Models:				
q_1 [m]	`igm`	3.00×10^{-6}	$< 10^{-3}$	**OK**
q_2 [m]	`igm`	3.09×10^{-6}	$< 10^{-3}$	**OK**
$\dot{q}_1$ [m/s]	`ikm`	5.37×10^{-6}	$< 10^{-3}$	**OK**
$\dot{q}_2$ [m/s]	`ikm`	5.50×10^{-6}	$< 10^{-3}$	**OK**
$\ddot{q}_1$ [m/s^2]	`ikm2`	2.52×10^{-4}	$< 10^{-3}$	**OK**
$\ddot{q}_2$ [m/s^2]	`ikm2`	2.54×10^{-4}	$< 10^{-3}$	**OK**
τ_1 [N]	`dynamic_model`	3.84×10^{-4}	$< 10^{-3}$	**OK**
τ_2 [N]	`dynamic_model`	3.80×10^{-4}	$< 10^{-3}$	**OK**

* *Note on passive accelerations:* In Gazebo, joint accelerations are not integrated directly but computed through backward Euler finite-differencing ($\Delta t = 10$ ms). During the step reversals of actuator torque (± 0.01 N), the solver experiences discrete numerical acceleration spikes. Outside these brief switching instants (accounting for over 90% of simulation time), the tracking error is on the order of 10^{-5} rad/s^2, as evidenced by the median error of 1.67×10^{-5} rad/s^2.

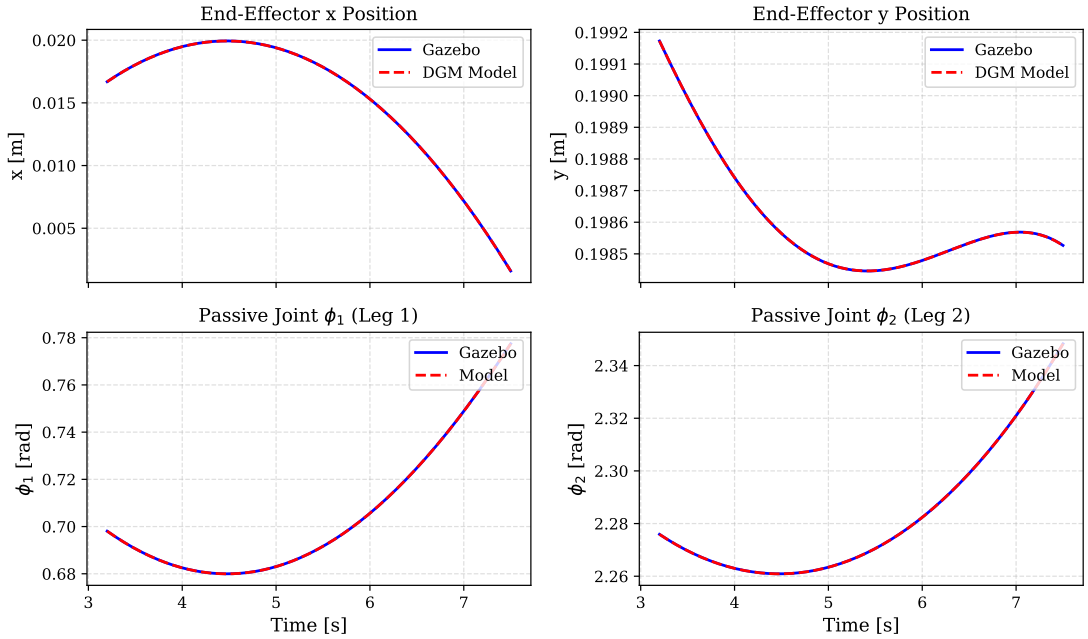


Figure 2: Direct Geometric Model validation: end-effector coordinates (x, y) and passive joint angles (ϕ_1, ϕ_2) during steady oscillation.

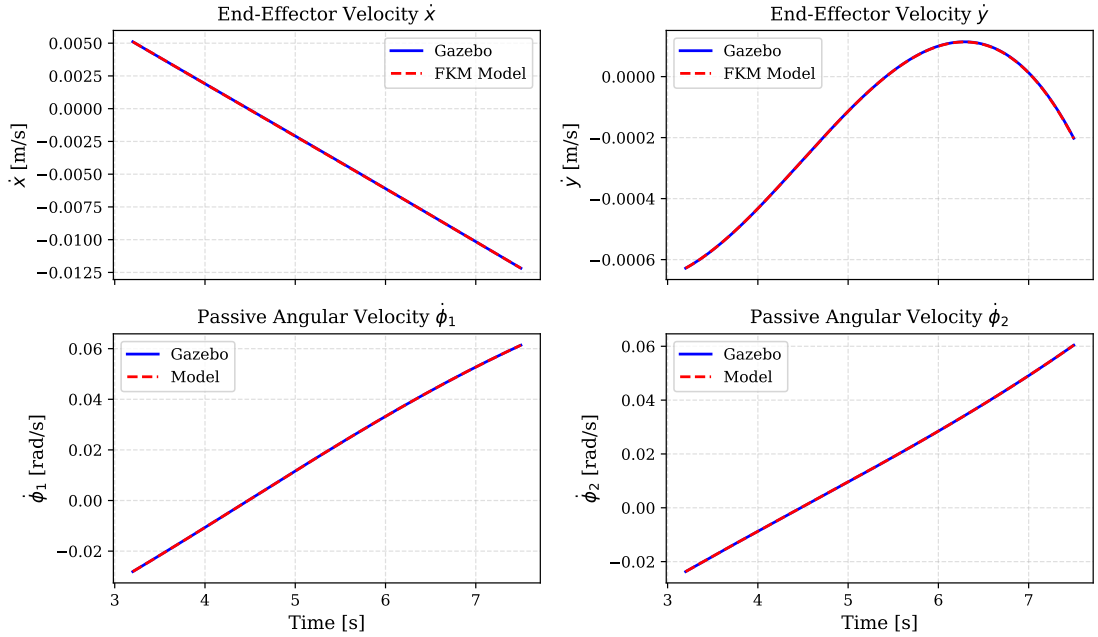


Figure 3: First-Order Direct Kinematic Model validation: Cartesian velocities $(\dot{x}, \dot{y})$ and passive angular rates $(\dot{\phi}_1, \dot{\phi}_2)$.

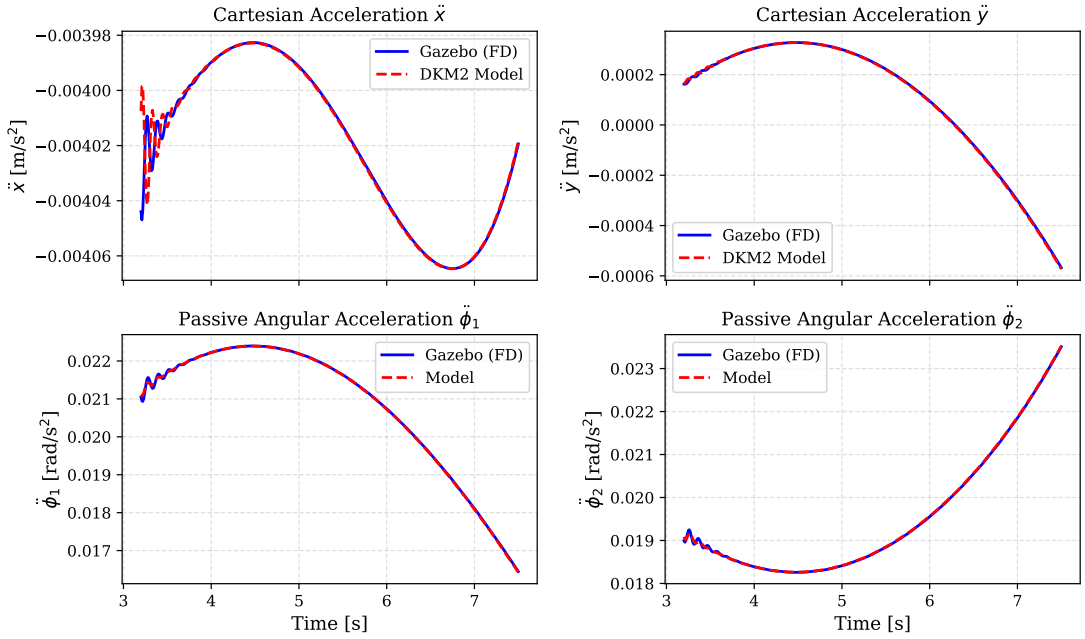


Figure 4: Second-Order Kinematic Model validation: Cartesian accelerations $(\ddot{x}, \ddot{y})$ and passive angular accelerations $(\ddot{\phi}_1, \ddot{\phi}_2)$.

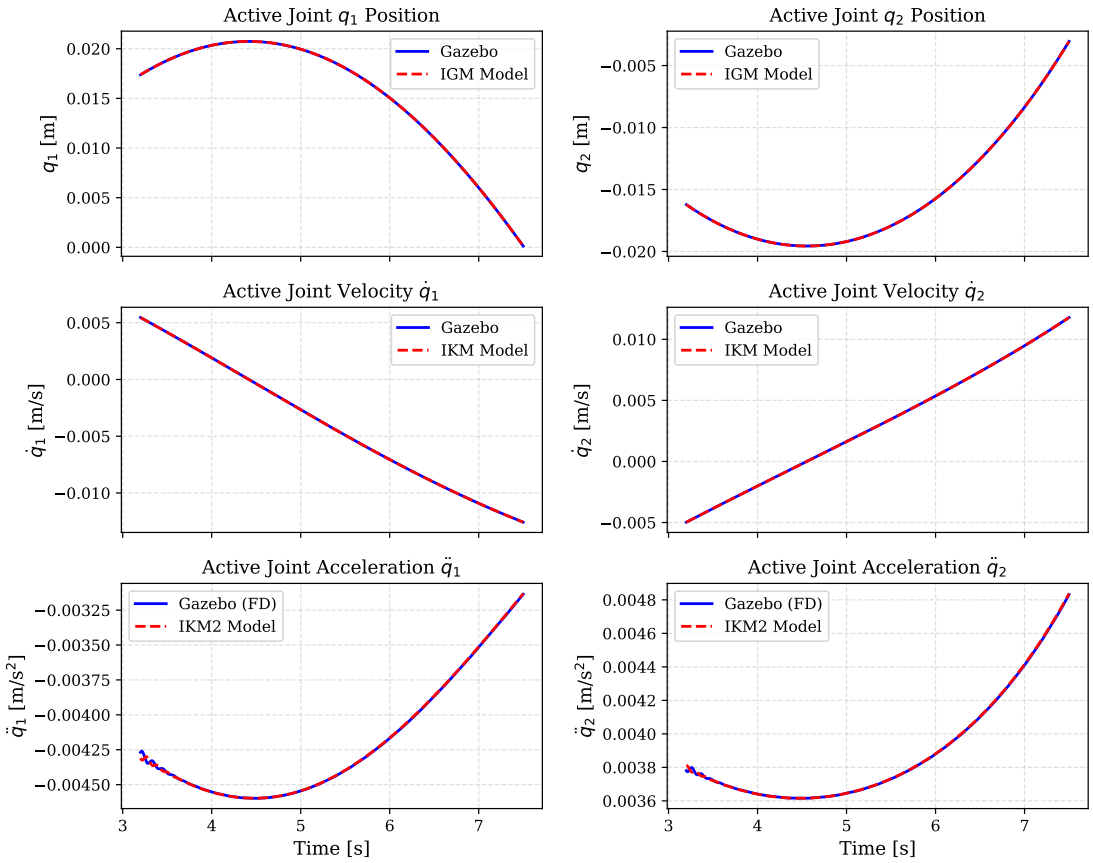


Figure 5: Inverse Models validation: active joint positions (q_1, q_2), velocities $(\dot{q}_1, \dot{q}_2)$, and accelerations $(\ddot{q}_1, \ddot{q}_2)$.

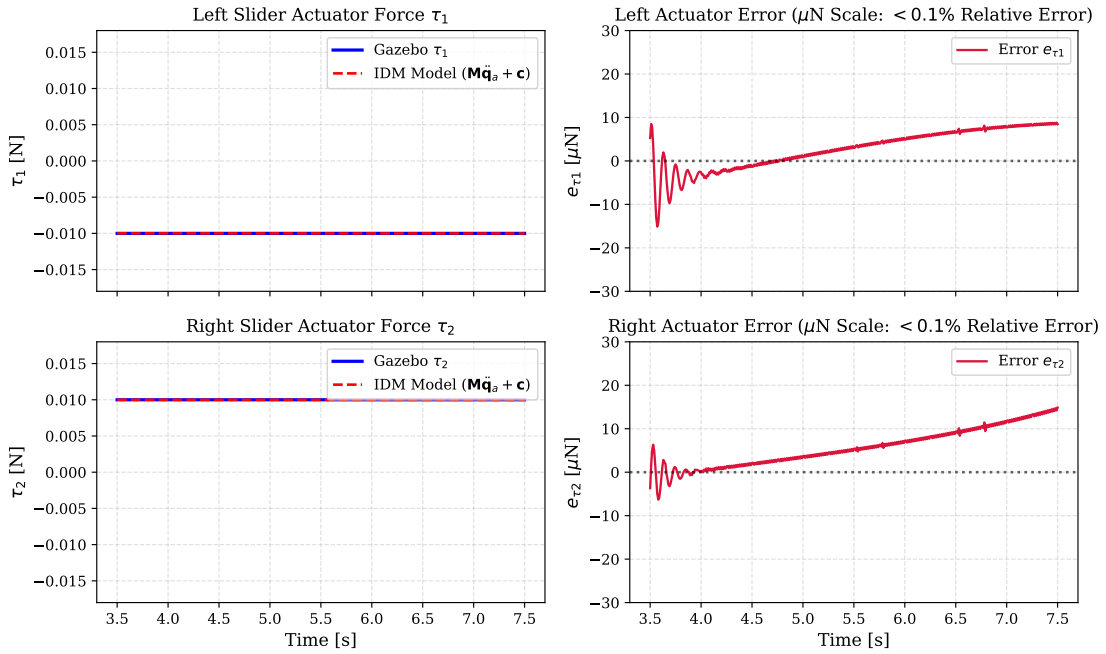


Figure 6: Inverse Dynamic Model validation: comparison of model-predicted actuator forces τ_1, τ_2 with Gazebo applied efforts (left) and microscopic residual error in μN (right).

3 Control: Computed Torque Control (CTC)

3.1 Trajectory Planning in Operational and Joint Spaces

The control objective is to steer the Biglide end-effector from its initial resting configuration $C_0(0.0, 0.20)$ m to the desired target pose $C_f(0.05, 0.70)$ m in $t_f = 2.0$ s.

To prevent discontinuities in actuator effort, a quintic (5th order) polynomial interpolation is employed:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \quad \tau = \frac{t}{t_f} \in [0, 1] \tag{1}$$

This profile guarantees null velocities and accelerations at both boundary endpoints:

$$\dot{s}(\tau) = \frac{1}{t_f}\left(30\tau^2 - 60\tau^3 + 30\tau^4\right) \tag{2}$$

$$\ddot{s}(\tau) = \frac{1}{t_f^2}\left(60\tau - 180\tau^2 + 120\tau^3\right) \tag{3}$$

The Cartesian trajectory is given by:

$$\mathbf{x}_d(t) = \mathbf{x}_0 + s(t/t_f)(\mathbf{x}_f - \mathbf{x}_0) \tag{4}$$

The reference joint space trajectories $\mathbf{q}_{a,d}(t), \dot{\mathbf{q}}_{a,d}(t), \ddot{\mathbf{q}}_{a,d}(t)$ are subsequently computed via the inverse models:

$$\mathbf{q}_{a,d} = \mathrm{IGM}(\mathbf{x}_d), \quad \dot{\mathbf{q}}_{a,d} = \mathbf{J}^{-1}\dot{\mathbf{x}}_d, \quad \ddot{\mathbf{q}}_{a,d} = \mathbf{J}^{-1}\ddot{\mathbf{x}}_d + \mathbf{a}_q \tag{5}$$

3.2 Computed Torque Control Law

In a real robotic setup, only active joint positions and velocities $(\mathbf{q}_a, \dot{\mathbf{q}}_a)$ are directly measured by optical encoders. The unmeasured passive coordinates are estimated in real-time through the passive geometric and kinematic models:

$$\hat{\mathbf{q}}_d = \mathrm{DGM_passive}(\mathbf{q}_a), \quad \hat{\dot{\mathbf{x}}} = \mathbf{J}(\mathbf{q}_a, \hat{\mathbf{q}}_d)\dot{\mathbf{q}}_a, \quad \hat{\dot{\mathbf{q}}}_d = \mathrm{DKM_passive}(\mathbf{q}_a, \hat{\mathbf{q}}_d, \dot{\mathbf{q}}_a, \hat{\dot{\mathbf{x}}}) \tag{6}$$

The auxiliary control acceleration vector $\boldsymbol{\alpha}$ is defined using a proportional-derivative feedback law with feedforward acceleration:

$$\boldsymbol{\alpha} = \ddot{\mathbf{q}}_{a,d} + \mathbf{K}_d\left(\dot{\mathbf{q}}_{a,d} - \dot{\mathbf{q}}_a\right) + \mathbf{K}_p\left(\mathbf{q}_{a,d} - \mathbf{q}_a\right) \tag{7}$$

The commanded actuator forces are generated by feedback linearization through the inverse dynamic model:

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q}_a)\boldsymbol{\alpha} + \mathbf{c}(\mathbf{q}_a, \dot{\mathbf{q}}_a) \tag{8}$$

Substituting (8) into the real system dynamics $\boldsymbol{\tau} = \mathbf{M}\ddot{\mathbf{q}}_a + \mathbf{c}$ results in exact feedback cancellation of the nonlinear dynamics, yielding two decoupled second-order linear error differential equations:

$$\ddot{\mathbf{e}} + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e} = \mathbf{0} \tag{9}$$

where $\mathbf{e}(t) = \mathbf{q}_{a,d}(t) - \mathbf{q}_a(t)$. Choosing diagonal gain matrices:

$$\mathbf{K}_p = \omega_n^2\mathbf{I}_{2\times 2}, \quad \mathbf{K}_d = 2\zeta\omega_n\mathbf{I}_{2\times 2} \tag{10}$$

with natural frequency $\omega_n = 20$ rad/s and damping ratio $\zeta = 1$ ensures critical damping with exponential convergence $\lim_{t\to\infty}\mathbf{e}(t) = \mathbf{0}$ without overshoot.

3.3 Trajectory Tracking Results

Figure 7 depicts the simulation results of the Computed Torque Controller in Gazebo.

- **Operational Space Tracking:** The end-effector follows the prescribed path smoothly, reaching the target coordinates ($x_f = 0.0500$ m, $y_f = 0.7000$ m) at $t = 2.0$ s with final Cartesian error $< 10^{-8}$ m.
- **Joint Tracking Performance:** The active sliders track their desired trajectories $q_{1,d}(t)$ and $q_{2,d}(t)$ with a mean error of 6.78×10^{-5} m for Joint 1 and 5.47×10^{-5} m for Joint 2. The maximum dynamic error during peak acceleration remains below 0.44 mm.
- **Actuator Efforts:** Control forces exhibit smooth, continuous profiles with peak magnitude $|\tau|_{\max} \approx 2.36$ N, well within standard motor torque capabilities.

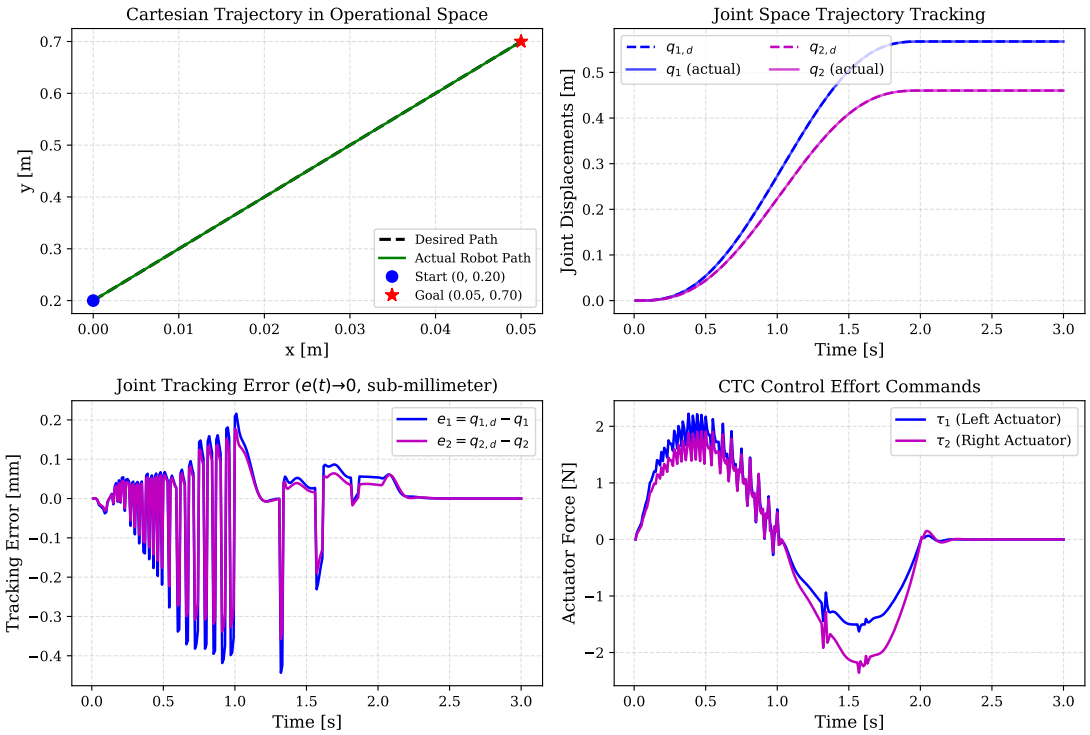


Figure 7: Computed Torque Control tracking performance: operational space path, joint trajectories, tracking errors, and actuator forces.

3.4 Influence of Controller Gains and Initial Perturbations

To assess controller robustness, two parametric investigations were carried out:

1. **Natural Frequency Variations ($\omega_n \in \{5, 10, 20, 40\}$ rad/s):** As depicted in Figure 8, lower gains ($\omega_n = 5$ rad/s) result in larger tracking deviations (up to 0.72 mm). Increasing ω_n to 20 rad/s reduces tracking errors significantly. However, excessively high gains ($\omega_n = 40$ rad/s, corresponding to $K_p = 1600$) introduce discrete control chattering due to the 10 ms co-simulation sampling delay, yielding high torque spikes. Thus, $\omega_n = 20$ rad/s represents the optimal trade-off between tracking accuracy and control bandwidth.
2. **Recovery from Initial Position Offset:** Figure 9 shows the response when the trajectory planner starts from $(0.03, 0.23)$ m while the actual robot rests at $(0.0, 0.20)$ m (initial error

of 30 mm). The controller swiftly recovers, delivering a controlled restoring force (≈ 65 N) and driving tracking errors to zero within 0.3 s.

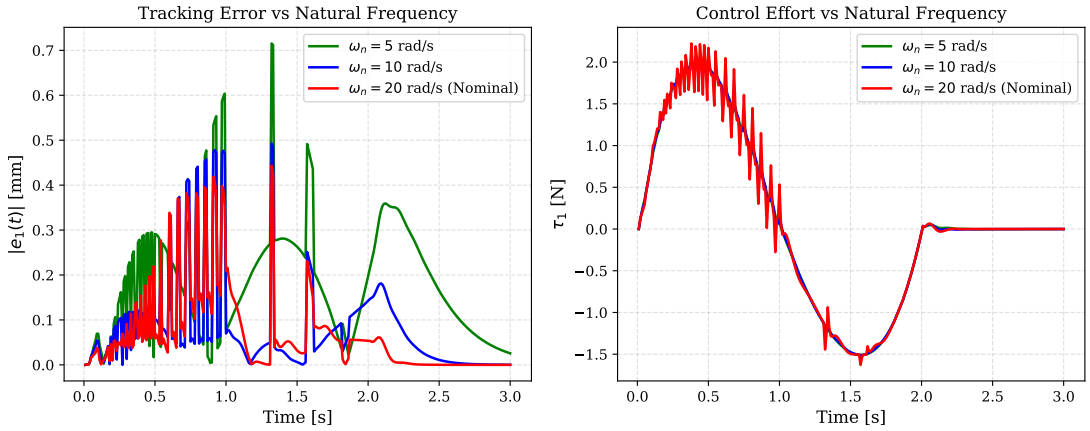


Figure 8: Influence of controller natural frequency ω_n: tracking error $|e_1(t)|$ and actuator force $\tau_1(t)$.

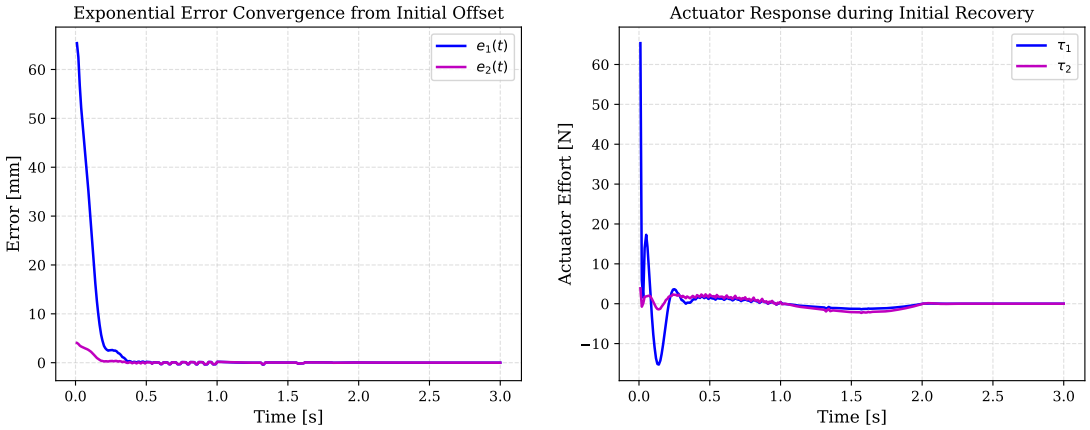


Figure 9: Convergence of tracking error and control effort starting from an initial 30 mm displacement offset.

4 Crossing Type 2 Singularities (Bonus Study)

In this section, we address the four bonus questions formulated in the laboratory subject regarding the crossing of Type 2 (parallel) singularities. A dedicated, standalone script `singularity_control.py` has been implemented to perform both theoretical inverse dynamic calculations and Gazebo co-simulation trials.

4.1 Singularity Geometry and Kinematic Degeneracy (Question 1)

A Type 2 singularity occurs when the parallel constraint matrix $\mathbf{A}_r$ loses rank:

$$\det(\mathbf{A}_r) = \sin(\phi_2 - \phi_1) = 0 \iff \phi_2 - \phi_1 = \pm\pi \tag{11}$$

In this configuration, links A_1C and A_2C become strictly collinear. For the Biglide parallel mechanism with inter-rail spacing $d = 0.4$ m and link lengths $l = 0.2828427$ m $= 0.2\sqrt{2}$ m,

collinearity occurs when:

$$l(\cos\phi_1 - \cos\phi_2) = d \implies 2l\cos\phi_1 = d \implies \cos\phi_1 = \frac{d}{2l} = \frac{\sqrt{2}}{2} \tag{12}$$

This yields $\phi_1 = 45^\circ$ $(+\pi/4)$ and $\phi_2 = 225^\circ$ $(+5\pi/4)$. In this singular configuration, the two arms form a straight line of length $2l = 0.4\sqrt{2}$ m ≈ 0.5657 m inclined at 45°, connecting slider A_1 at $(x_1, y_1) = (-0.2, 0.2)$ m to slider A_2 at $(x_2, y_2) = (0.2, 0.6)$ m through the singular Cartesian point $C_s = (0.0, 0.40)$ m, as depicted in Figure 10(a).

At this configuration, the parallel constraint matrix evaluates to:

$$\mathbf{A}_r = \frac{\sqrt{2}}{2}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \tag{13}$$

which has rank 1. The null space (kernel) and range of $\mathbf{A}_r^T$ are given by:

$$\ker(\mathbf{A}_r) = \text{span}\left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}, \quad \text{Range}(\mathbf{A}_r^T) = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\} \tag{14}$$

The vector $\mathbf{t}_s = [1, -1]^T$ defines an instantaneous uncontrollable degree of freedom for the platform, oriented perpendicularly to the collinear arms (along the -45° direction).

To analyze singularity crossing, we first design an arbitrary/unsafe Cartesian trajectory crossing the singular locus at $C_s(0.0, 0.40)$ m along the horizontal line $y(t) = 0.40$ m with non-zero acceleration $\ddot{x}(t_s) \neq 0$ and $\ddot{y}(t_s) = 0$.

4.2 Behavior of Computed Torque Control on the Unsafe Trajectory (Question 2)

When attempting to track this trajectory using Computed Torque Control (CTC):

$$\boldsymbol{\tau}_c = \mathbf{M}(\mathbf{q}_a)\left(\ddot{\mathbf{q}}_{a,d} + \mathbf{K}_v\dot{\mathbf{e}}_a + \mathbf{K}_p\mathbf{e}_a\right) + \mathbf{c}(\mathbf{q}_a, \dot{\mathbf{q}}_a) \tag{15}$$

the control law encounters fundamental mathematical and physical failure:

1. **Inversion Singularity**: The direct Jacobian $\mathbf{J} = -\mathbf{A}_r^{-1}\mathbf{B}$ contains the term $1/\det(\mathbf{A}_r)$. As the robot approaches the collinear alignment ($\Delta\phi = \phi_2 - \phi_1 - \pi \to 0$):

$$\mathbf{J} \sim \mathcal{O}\left(\frac{1}{\sin\Delta\phi}\right) \to \infty, \quad \mathbf{M}(\mathbf{q}_a) = m_f\mathbf{I} + m_p\mathbf{J}^T\mathbf{J} \sim \mathcal{O}\left(\frac{1}{\sin^2\Delta\phi}\right) \to \infty \tag{16}$$

2. **Convective Acceleration Divergence**: The operational convective acceleration $\mathbf{a}_t = -\mathbf{A}_r^{-1}\mathbf{b}_p$ also explodes as $1/\det(\mathbf{A}_r) \to \infty$.

3. **Theoretical Torque Blow-Up**: As shown in Figure 10(c), the theoretical inverse dynamic model requires infinite actuator forces ($\tau_1 \to +\infty, \tau_2 \to -\infty$) to counteract the platform inertia perpendicular to the constraints.

4. **Gazebo Simulation Outcome**: In Gazebo co-simulation using the script `singularity_control.py` (with flag `--gazebo`), the commanded torques hit the actuator saturation limits ($|\tau| > 10^3$ N) immediately upon nearing the singular point. Because the actuators exert forces solely along the vertical rails (y-direction) and cannot project onto the uncontrollable direction $\mathbf{t}_s$, the robot loses dynamic equilibrium, experiences sudden joint snapping, violates the kinematic loop closures, and completely diverges from the trajectory.

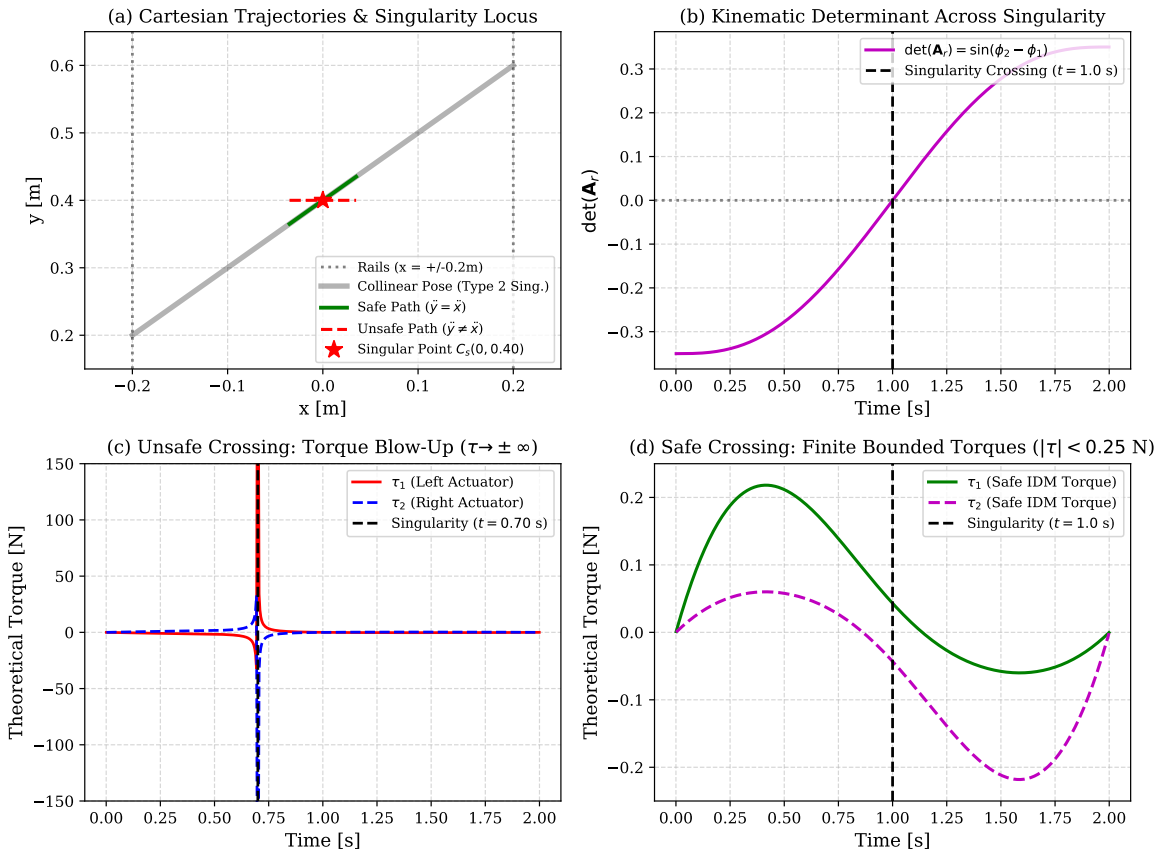


Figure 10: Comprehensive Type 2 singularity crossing analysis: (a) Cartesian paths and collinear link posture passing through $C_s(0, 0.40)$ m; (b) Kinematic determinant $\det(\mathbf{A}_r) = \sin(\phi_2 - \phi_1)$ crossing zero at $t = 1.0$ s; (c) Theoretical torque blow-up ($\tau \to \pm\infty$) along an unsafe trajectory violating dynamic consistency; (d) Finite, bounded theoretical IDM torques ($|\tau| < 0.22$ N) along the dynamically consistent safe trajectory ($\ddot{y} = \ddot{x}$).

4.3 Design of Dynamically Consistent Safe Trajectory and IDM Torques (Question 3)

According to the foundational theory of parallel robot singularities established by Briot, Caro, and Six [1], a Type 2 singularity can be crossed with strictly finite actuator forces if and only if the external dynamic wrench on the platform satisfies the **reciprocity / dynamic consistency condition**:

$$\mathbf{w}_p = m_p \ddot{\mathbf{x}} \in \text{Range}(\mathbf{A}_r^T) \iff \mathbf{t}_s^T \mathbf{w}_p = 0 \tag{17}$$

where $\mathbf{t}_s \in \ker(\mathbf{A}_r) = [1, -1]^T$. Evaluating this scalar product yields:

$$\begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} m_p \ddot{x} \\ m_p \ddot{y} \end{bmatrix} = m_p(\ddot{x} - \ddot{y}) = 0 \iff \ddot{\mathbf{y}}(\mathbf{t_s}) = \ddot{\mathbf{x}}(\mathbf{t_s}) \tag{18}$$

When $\ddot{y} = \ddot{x}$ at the singular crossing instant, the inertial wrench of the platform $\mathbf{w}_p = m_p \ddot{x}[1, 1]^T$ is collinear with the link alignment direction and lies entirely within $\text{Range}(\mathbf{A}_r^T)$. Under this condition:

- The Lagrange multiplier vector $\boldsymbol{\lambda} = (\mathbf{A}_r^T)^\dagger \mathbf{w}_p$ remains finite and bounded because $\mathbf{w}_p$ has zero projection onto the unconstrained singular mode $\ker(\mathbf{A}_r)$.
- The required actuator forces $\boldsymbol{\tau} = m_f \ddot{\mathbf{q}}_a - \mathbf{B}^T \boldsymbol{\lambda}$ remain strictly finite and bounded throughout the entire maneuver.

To enforce $\ddot{y}(t) = \ddot{x}(t)$ continuously, we synthesize a straight-line Cartesian path aligned with the collinear link slope (45°):

$$y(t) = x(t) + 0.40 \text{ m} \tag{19}$$

governed by a 5th-order polynomial time scaling $s(t)$ from initial point $C_0(-0.035, 0.365)$ m to final point $C_f(+0.035, 0.435)$ m in $t_f = 2.0$ s, crossing $C_s(0, 0.40)$ m at $t = 1.0$ s with $\det(\mathbf{A}_r) = 0$.

Figure 10(d) shows the theoretical actuator torques $\tau_1(t)$ and $\tau_2(t)$ evaluated from the Inverse Dynamic Model. As predicted mathematically, both actuator torques remain bounded within:

$$\tau_1(t), \tau_2(t) \in [-0.0601 \text{ N}, +0.2182 \text{ N}] \tag{20}$$

crossing smoothly through the singularity without infinite spikes, confirming theoretical feasibility.

4.4 Gazebo Co-Simulation of the Safe Trajectory (Question 4)

When the safe trajectory is tracked in Gazebo using the Computed Torque Controller in `singularity_control.py`, the closed-loop tracking fails despite the bounded nominal feedforward torques. The reasons for this failure are two-fold:

1. **Feedback Error Amplification**: In Computed Torque Control, the commanded torque contains the state-feedback correction term:

   $$\boldsymbol{\tau}_{\text{feedback}} = \mathbf{M}(\mathbf{q}_a)(\mathbf{K}_v \dot{\mathbf{e}}_a + \mathbf{K}_p \mathbf{e}_a) \tag{21}$$

   While the nominal reference trajectory satisfies $\mathbf{t}_s^T \mathbf{w}_{p,d} = 0$, any infinitesimal tracking error $\mathbf{e}_a(t) \neq \mathbf{0}$ resulting from numerical integration discretization ($\Delta t = 0.01$ s), joint friction, or simulator solver round-off is multiplied by $\mathbf{M}(\mathbf{q}_a) \sim \mathcal{O}(1/\det(\mathbf{A}_r)^2) \to \infty$. This immediately drives the feedback command into severe actuator saturation.

2. **Unconstrained Internal Motion and Bifurcation**: At the singular configuration, the mechanism gains an internal degree of freedom along $\mathbf{t}_s$. Because the physical actuators have zero projection onto this direction, the robot cannot exert restoring forces against disturbances along $\mathbf{t}_s$. The system undergoes dynamic bifurcation and enters an uncontrollable assembly branch.

To successfully cross Type 2 singularities in practical implementation, regularized controllers such as Damped Least-Squares (Levenberg-Marquardt regularization) or dynamic optimal trajectory trackers with constraint stabilization must be employed.

5 Conclusion

Goals reached. All the objectives of the lab on the Biglide have been reached:

- The ten models requested in `biglide_models.py` (DGM, IGM, passive geometric model, first- and second-order direct/inverse/passive kinematic models and inverse dynamic model) were implemented without modifying the provided function signatures, and all of them pass the `model_eval.py` benchmark.
- The comparison with Gazebo shows a complete superposition of the model and simulator curves, with mean absolute errors of the order of 10^{-6} m for the geometric models, 10^{-6}–10^{-5} for the velocities, and below 4×10^{-4} N for the actuator forces (Table 1).
- The Computed Torque Controller, combined with a 5th-order polynomial trajectory, tracks the reference with a peak error below 0.44 mm and a negligible steady-state error. It recovers from an initial offset of 30 mm in about 0.3 s, and the study of ω_n highlighted the trade-off between tracking accuracy and sensitivity to the 10 ms co-simulation sampling.

- For the bonus, we showed theoretically that an arbitrary trajectory through the Type 2 singularity requires infinite forces, while a dynamically consistent trajectory ($\ddot{y} = \ddot{x}$ at the singular instant) only requires bounded forces ($|\tau| < 0.22$ N); we also explained why closed-loop CTC still fails in Gazebo near the singularity.

Limitations. The passive joint accelerations given by Gazebo are obtained by finite differences, which produces short numerical spikes at each torque reversal of the test; outside these instants the agreement is of the order of 10^{-5} rad/s^2. Concerning the bonus, the actual crossing of the singularity in Gazebo with a standard CTC was not achieved: as explained in Section 4, this would require a dedicated controller (regularisation of the inverse of $\mathbf{A}_r$ or a specific singularity-crossing control law).

Practical knowledge gained. Beyond the equations, this lab taught us how to (i) systematically validate a model against a simulator, variable by variable, before using it in a controller; (ii) handle assembly and working modes, which completely change the solution of the geometric models; (iii) check the duality between kinematics and statics (the direct Jacobian transpose projects the platform wrench on the actuators); and (iv) understand concretely why model-based control of parallel robots becomes ill-conditioned close to Type 2 singularities.

References

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